

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline

data = pd.read_csv('FDI data.csv')

# First five rows of dataset
data.head()
```

	Sector	2000-01	2001-02	2002-03	2003-04	2004-
05 \						
0	METALLURGICAL INDUSTRIES	22.69	14.14	36.61	8.11	
200.38						
1	MINING	1.32	6.52	10.06	23.48	
9.92						
2	POWER	89.42	757.44	59.11	27.09	
43.37						
3	NON-CONVENTIONAL ENERGY	0.00	0.00	1.70	4.14	
1.27						
4	COAL PRODUCTION	0.00	0.00	0.00	0.04	
0.00						

	2005-06	2006-07	2007-08	2008-09	2009-10	2010-11	2011-12
2012-13 \							
0	149.13	169.94	1175.75	959.94	419.88	1098.14	1786.14
1466.23							
1	7.40	6.62	444.36	34.16	174.40	79.51	142.65
57.89							
2	72.69	157.15	988.68	907.66	1271.79	1271.77	1652.38
535.68							
3	1.35	2.44	58.82	125.88	622.52	214.40	452.17
1106.52							
4	9.14	1.30	14.08	0.22	0.00	0.00	0.00
0.00							

	2013-14	2014-15	2015-16	2016-17
0	567.63	359.34	456.31	1440.18
1	12.73	684.39	520.67	55.75
2	1066.08	707.04	868.80	1112.98
3	414.25	615.95	776.51	783.57
4	2.96	0.00	0.00	0.00

```
#Last five rows of dataset
data.tail()
```

	Sector	2000-01	2001-
02 \			
58	PRINTING OF BOOKS (INCLUDING LITHO PRINTING IN...	0.00	
0.00			

59	COIR	0.00
0.00		
60	CONSTRUCTION (INFRASTRUCTURE) ACTIVITIES	0.00
0.00		
61	CONSTRUCTION DEVELOPMENT: Townships, housing, ...	24.33
51.75		
62	MISCELLANEOUS INDUSTRIES	832.07
221.37		

	2002-03	2003-04	2004-05	2005-06	2006-07	2007-08	2008-09
2009-10 \							
58	6.30	0.00	0.06	9.90	20.04	35.54	31.61
70.51							
59	0.00	0.00	0.47	0.59	0.04	0.01	0.00
0.25							
60	0.00	0.00	0.00	0.93	64.06	182.92	172.70
324.56							
61	36.10	47.04	152.06	228.71	1392.95	3887.33	4657.51
5466.13							
62	218.76	235.48	121.83	164.76	304.87	528.42	1549.70
1147.56							

	2010-11	2011-12	2012-13	2013-14	2014-15	2015-16	2016-17
58	36.63	47.39	14.34	113.78	72.58	122.81	53.17
59	0.10	0.55	0.15	0.54	1.36	0.00	0.00
60	675.07	386.28	283.89	485.37	870.25	4510.71	1860.73
61	1663.03	3140.78	1332.49	1226.05	769.14	112.55	105.14
62	1475.97	813.38	229.49	468.74	765.88	668.77	296.40

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 63 entries, 0 to 62
Data columns (total 18 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Sector      63 non-null    object
1   2000-01     63 non-null    float64
2   2001-02     63 non-null    float64
3   2002-03     63 non-null    float64
4   2003-04     63 non-null    float64
5   2004-05     63 non-null    float64
6   2005-06     63 non-null    float64
7   2006-07     63 non-null    float64
8   2007-08     63 non-null    float64
9   2008-09     63 non-null    float64
10  2009-10     63 non-null    float64
11  2010-11     63 non-null    float64
12  2011-12     63 non-null    float64
13  2012-13     63 non-null    float64
```

```

14 2013-14 63 non-null float64
15 2014-15 63 non-null float64
16 2015-16 63 non-null float64
17 2016-17 63 non-null float64

```

```
dtypes: float64(17), object(1)
```

```
memory usage: 9.0+ KB
```

```
data.describe()
```

	2000-01	2001-02	2002-03	2003-04	2004-05 \
count	63.000000	63.000000	63.000000	63.000000	63.000000
mean	37.757302	63.931587	42.925714	34.727778	51.090317
std	112.227860	157.878737	86.606439	67.653735	101.934873
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	0.200000	0.215000	0.715000
50%	4.030000	5.070000	11.010000	6.370000	9.090000
75%	23.510000	44.830000	36.555000	38.660000	43.205000
max	832.070000	873.230000	419.960000	368.320000	527.900000

	2005-06	2006-07	2007-08	2008-09	2009-10 \
count	63.000000	63.000000	63.000000	63.000000	63.000000
mean	87.932540	198.281905	390.085714	498.348571	410.069524
std	206.436967	686.783115	1026.249935	1134.649040	926.814626
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.230000	4.160000	9.950000	11.950000	7.880000
50%	22.620000	25.820000	58.820000	84.880000	69.740000
75%	63.855000	108.325000	279.270000	383.320000	341.595000
max	1359.970000	4713.780000	6986.170000	6183.490000	5466.130000

	2010-11	2011-12	2012-13	2013-14	2014-15 \
count	63.000000	63.000000	63.000000	63.000000	63.000000
mean	339.413810	557.472698	355.930000	385.703492	490.959841
std	627.141139	1031.474056	778.091368	658.429944	837.787060
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	8.430000	22.720000	15.115000	16.610000	33.800000
50%	58.070000	129.360000	95.410000	113.780000	177.220000

75%	304.280000	593.525000	288.025000	473.060000	595.390000
max	3296.090000	5215.980000	4832.980000	3982.890000	4443.260000

	2015-16	2016-17
count	63.000000	63.000000
mean	634.936349	690.131111
std	1335.307706	1411.965354
min	0.000000	0.000000
25%	30.000000	19.905000
50%	159.130000	110.860000
75%	519.070000	741.220000
max	6889.460000	8684.070000

Get number of rows and columns from the dataset

data.shape

data['Sector'][36]

'FERMENTATION INDUSTRIES'

#Names of the required columns

data.columns

```
Index(['Sector', '2000-01', '2001-02', '2002-03', '2003-04', '2004-05',
      '2005-06', '2006-07', '2007-08', '2008-09', '2009-10', '2010-11',
      '2011-12', '2012-13', '2013-14', '2014-15', '2015-16', '2016-17'],
      dtype='object')
```

Check for values which are null if any

pd.isnull(data).sum

```
<bound method DataFrame.sum of      Sector  2000-01  2001-02  2002-03
2003-04  2004-05  2005-06  2006-07 \
0   False   False   False   False   False   False   False
False
1   False   False   False   False   False   False   False
False
2   False   False   False   False   False   False   False
False
3   False   False   False   False   False   False   False
False
4   False   False   False   False   False   False   False
False
...      ...      ...      ...      ...      ...      ...      ...
...
58   False   False   False   False   False   False   False
```

```

False
59  False      False      False      False      False      False      False      False
False
60  False      False      False      False      False      False      False      False
False
61  False      False      False      False      False      False      False      False
False
62  False      False      False      False      False      False      False      False
False

      2007-08  2008-09  2009-10  2010-11  2011-12  2012-13  2013-14
2014-15 \
0      False      False      False      False      False      False      False
False
1      False      False      False      False      False      False      False
False
2      False      False      False      False      False      False      False
False
3      False      False      False      False      False      False      False
False
4      False      False      False      False      False      False      False
False
..      ...      ...      ...      ...      ...      ...      ...
...
58      False      False      False      False      False      False      False
False
59      False      False      False      False      False      False      False
False
60      False      False      False      False      False      False      False
False
61      False      False      False      False      False      False      False
False
62      False      False      False      False      False      False      False
False

      2015-16  2016-17
0      False      False
1      False      False
2      False      False
3      False      False
4      False      False
..      ...      ...
58      False      False
59      False      False
60      False      False
61      False      False
62      False      False

```

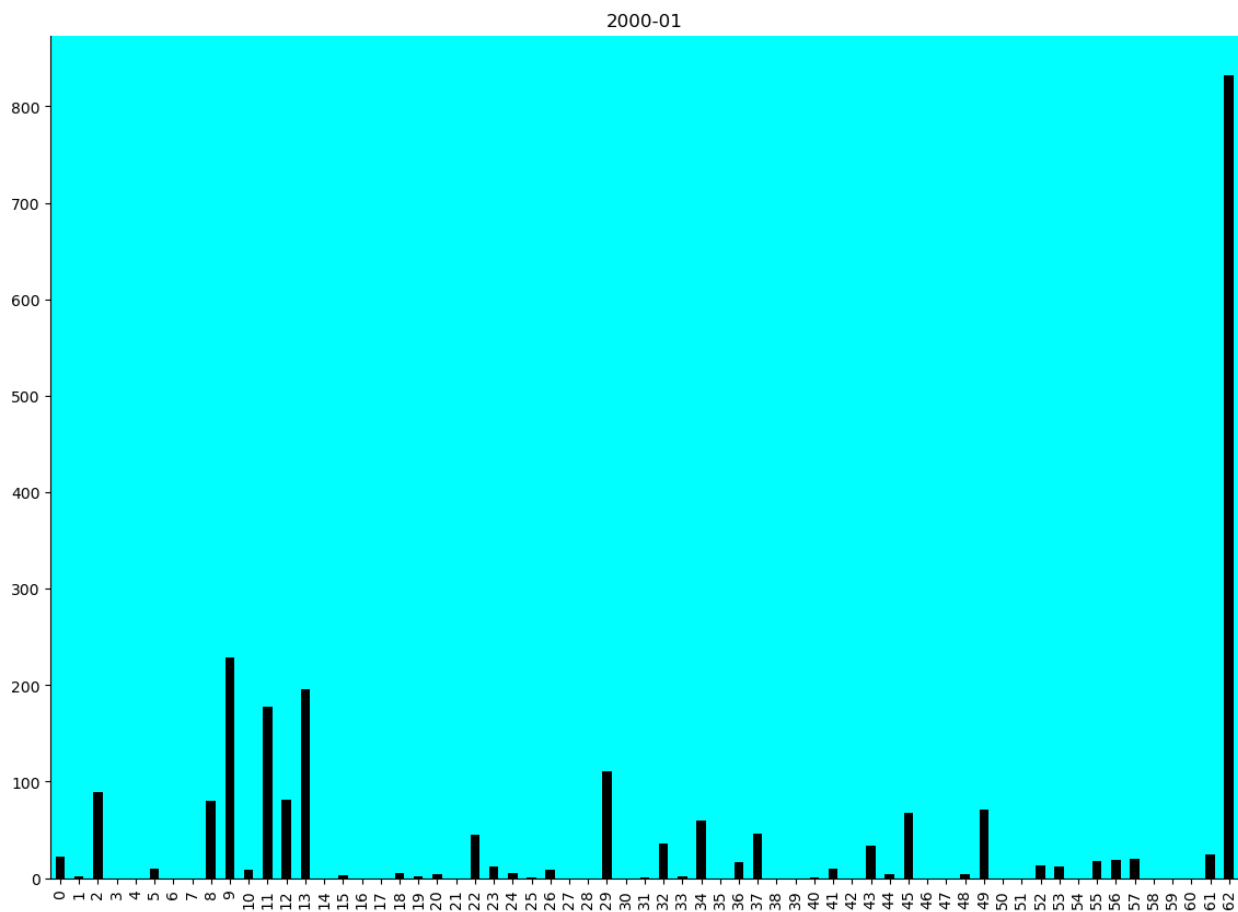
```
[63 rows x 18 columns]>
```

```
#Check for duplicate values
data.duplicated().value_counts()
```

```
False      63
Name: count, dtype: int64
```

```
# Analysis of sectors with highest investment in the first financial year
```

```
ax = plt.axes()
ax.set_facecolor('cyan')
data['2000-01'].plot(kind='bar',figsize=(14,10),title='2000-01',color='black')
plt.gca().spines[['top','right']].set_visible(False)
```

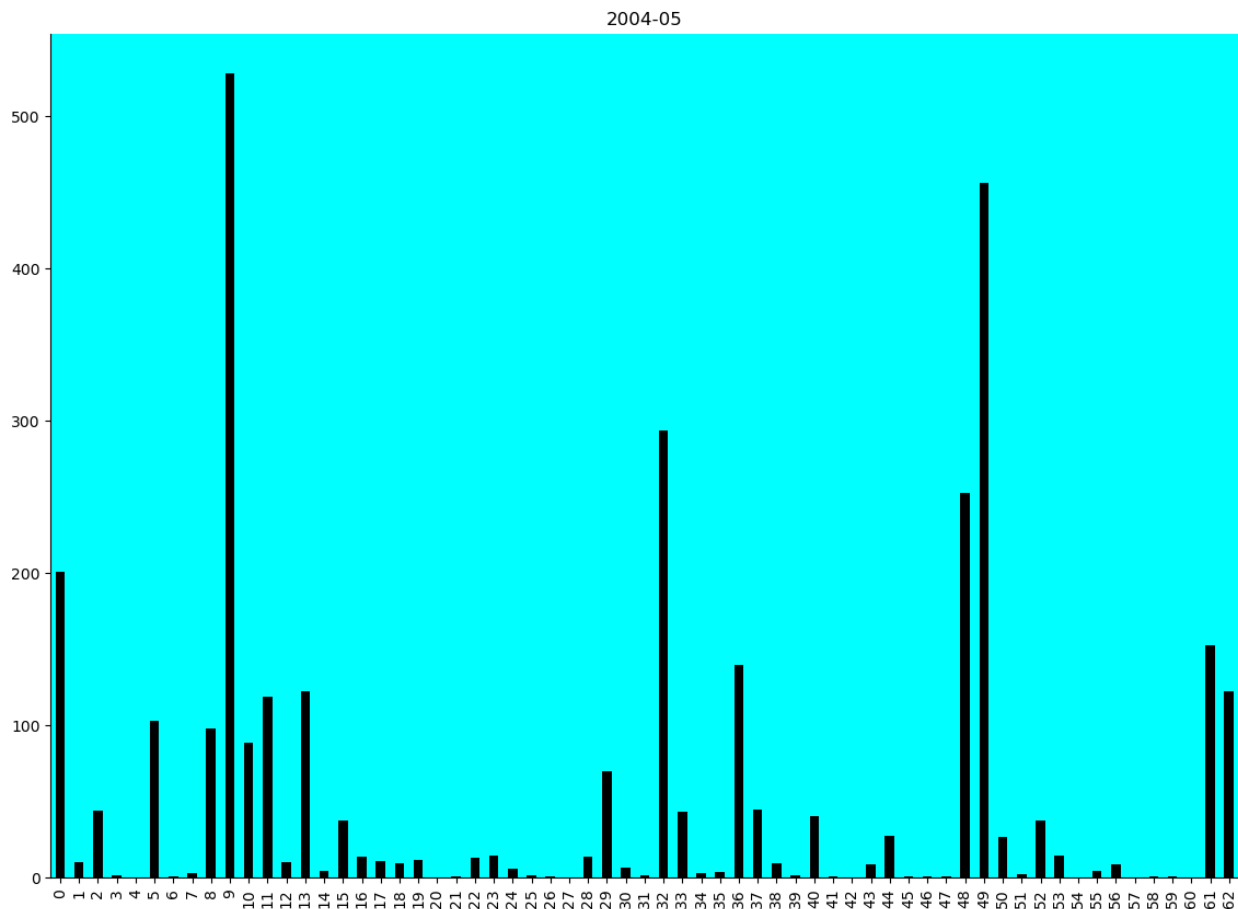


```
# Top 5 industries in first financial year with most investment
```

```
print(data['Sector'][62])
print(data['Sector'][9])
print(data['Sector'][13])
print(data['Sector'][11])
print(data['Sector'][29])
```

MISCELLANEOUS INDUSTRIES
 COMPUTER SOFTWARE & HARDWARE
 AUTOMOBILE INDUSTRY
 TELECOMMUNICATIONS
 CHEMICALS (OTHER THAN FERTILIZERS)

```
# Analysis of same sectors after 4 years
ax = plt.axes()
ax.set_facecolor('cyan')
data['2004-05'].plot(kind='bar',figsize=(14,10),title='2004-05',color='black')
plt.gca().spines[['top','right']].set_visible(False)
```



```
# Top 5 industries with most financial investment in year 2004-05
print(data['Sector'][9])
print(data['Sector'][49])
print(data['Sector'][32])
print(data['Sector'][48])
print(data['Sector'][0])
```

COMPUTER SOFTWARE & HARDWARE
 SERVICES SECTOR (Fin.,Banking,Insurance,Non

```

Fin/Business,Outsourcing,R&D,Courier,Tech. Testing and Analysis,
Other)
DRUGS & PHARMACEUTICALS
CONSULTANCY SERVICES
METALLURGICAL INDUSTRIES

```

```

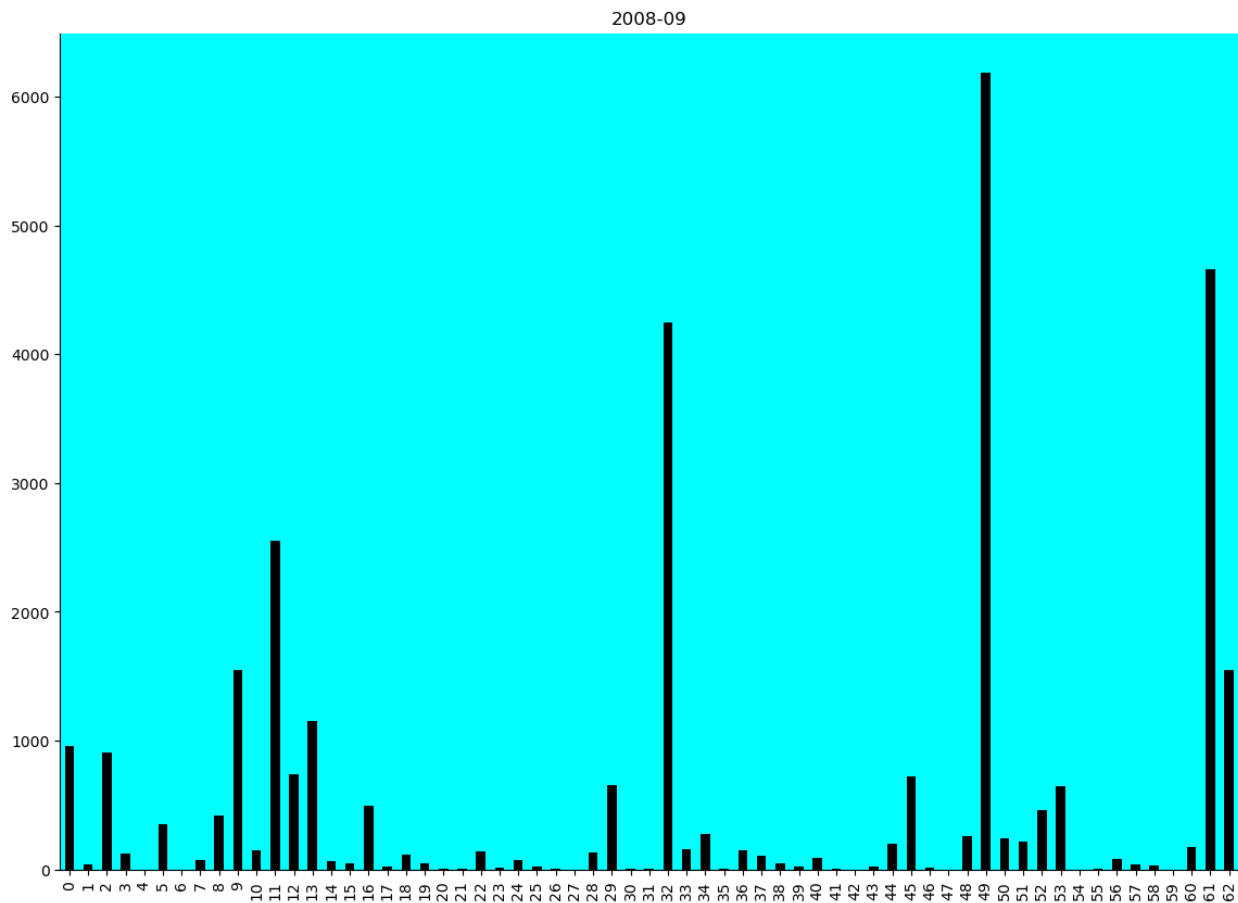
# Analysis of same sectors after another 4 years

```

```

ax = plt.axes()
ax.set_facecolor('cyan')
data['2008-09'].plot(kind='bar',figsize=(14,10),title='2008-
09',color='black')
plt.gca().spines[['top','right']].set_visible(False)

```



```

# Top 5 industries with most financial investment in year 2008-09

```

```

print(data['Sector'][49])
print(data['Sector'][61])
print(data['Sector'][32])
print(data['Sector'][11])
print(data['Sector'][9])

```

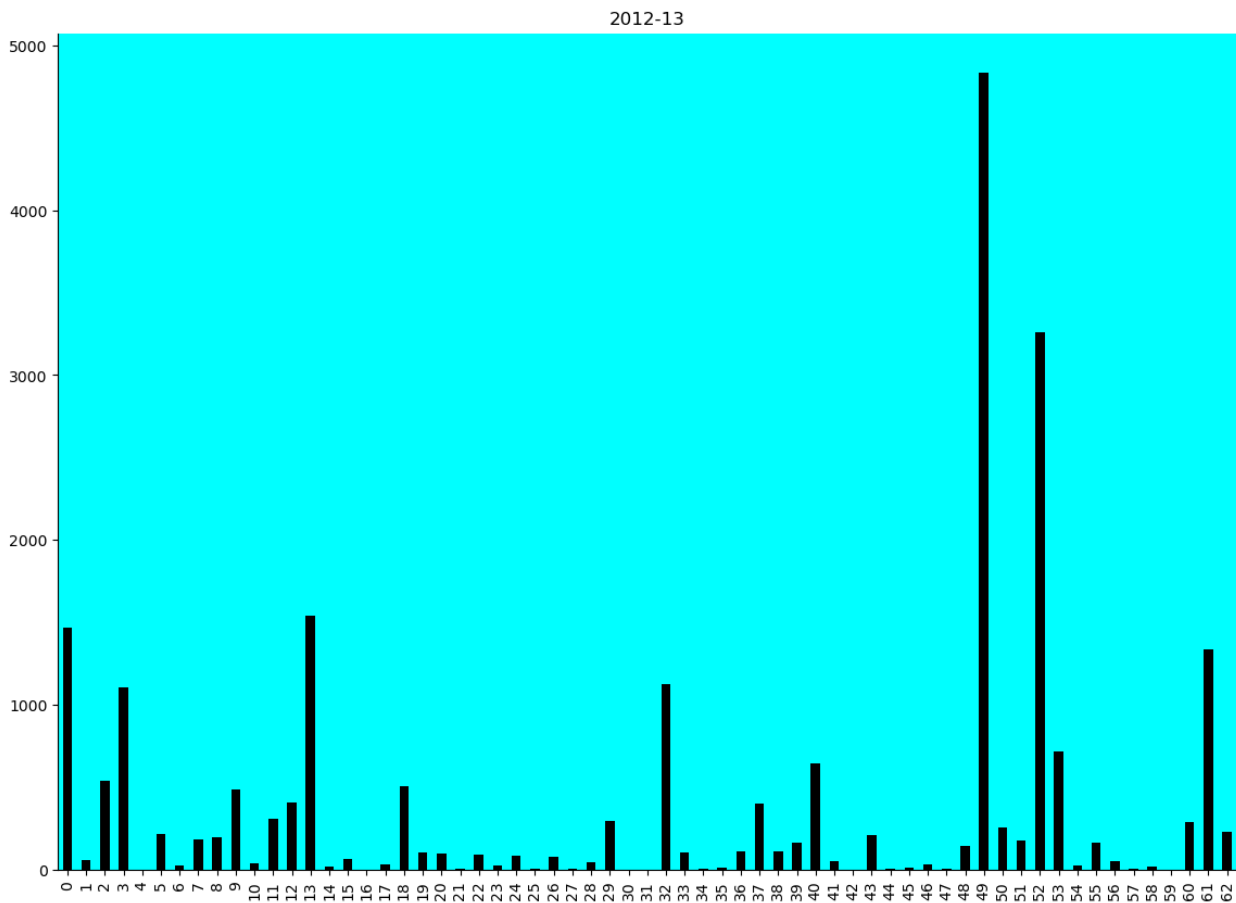
```

SERVICES SECTOR (Fin.,Banking,Insurance,Non
Fin/Business,Outsourcing,R&D,Courier,Tech. Testing and Analysis,

```

```
Other)
CONSTRUCTION DEVELOPMENT: Townships, housing, built-up infrastructure
and construction-development projects
DRUGS & PHARMACEUTICALS
TELECOMMUNICATIONS
COMPUTER SOFTWARE & HARDWARE
```

```
ax = plt.axes()
ax.set_facecolor('cyan')
data['2012-13'].plot(kind='bar',figsize=(14,10),title='2012-
13',color='black')
plt.gca().spines[['top','right']].set_visible(False)
```



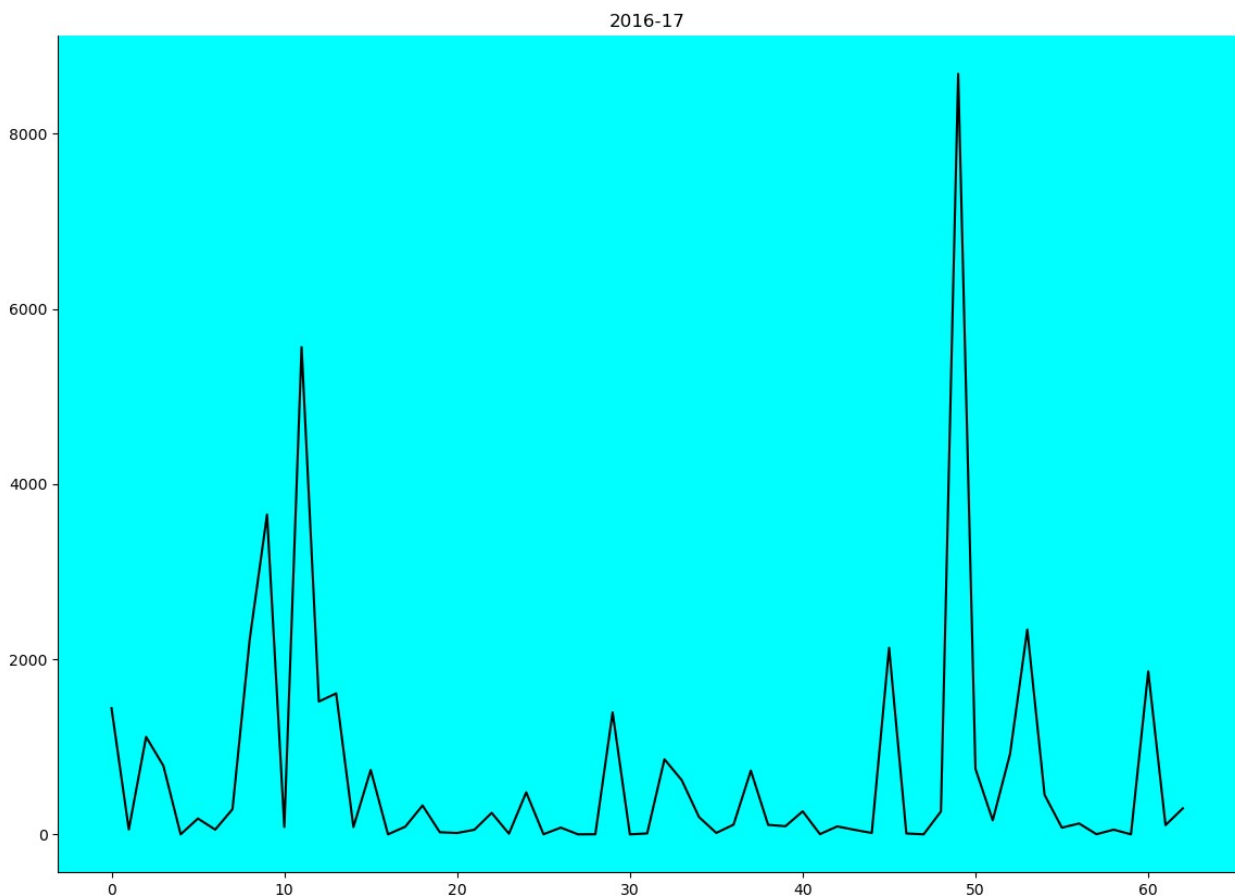
```
# Top 5 industries with most financial investment in year 2012-13
print(data['Sector'][49])
print(data['Sector'][52])
print(data['Sector'][13])
print(data['Sector'][0])
print(data['Sector'][61])
```

```
SERVICES SECTOR (Fin.,Banking,Insurance,Non
Fin/Business,Outsourcing,R&D,Courier,Tech. Testing and Analysis,
```

```
Other)
HOTEL & TOURISM
AUTOMOBILE INDUSTRY
METALLURGICAL INDUSTRIES
CONSTRUCTION DEVELOPMENT: Townships, housing, built-up infrastructure
and construction-development projects
```

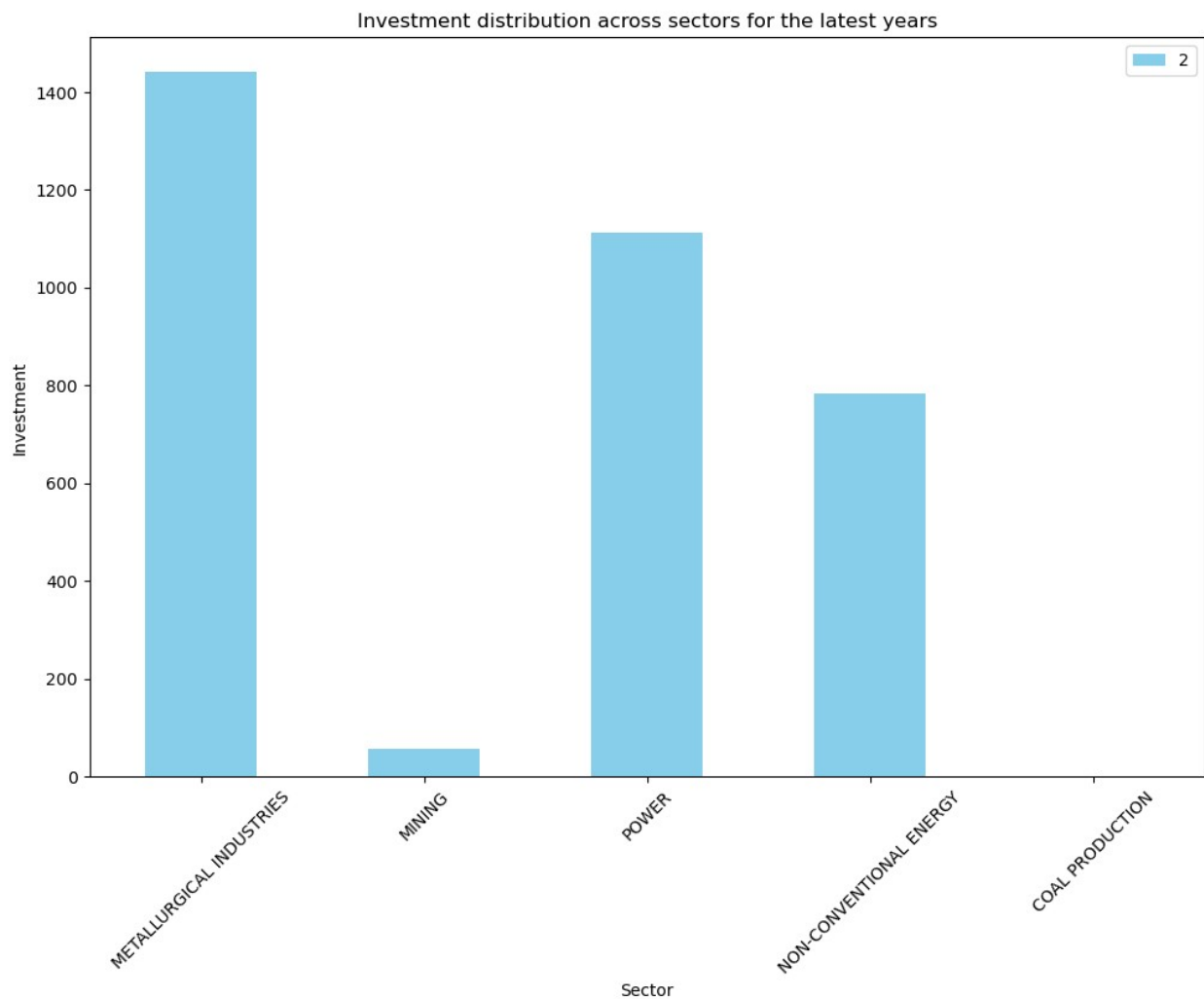
```
# Analysis of investments in sectors from 2016-17
```

```
ax = plt.axes()
ax.set_facecolor('cyan')
data['2016-17'].plot(kind='line',figsize=(14,10),title='2016-
17',color='black')
plt.gca().spines[['top','right']].set_visible(False)
```



```
latest_year = data.columns[-1]
investment_latest_year = data[[latest_year,'Sector']].head()
investment_latest_year = investment_latest_year.set_index('Sector')
investment_latest_year.plot(kind='bar',figsize=(12,8),color='skyblue')
plt.title('Investment distribution across sectors for the latest
years')
plt.xlabel('Sector')
plt.ylabel('Investment')
```

```
plt.xticks(rotation=45)
plt.legend(latest_year)
plt.show()
```



```
corr_matrix = data.iloc[:,1:].corr()
plt.figure(figsize=(10,8))
sns.heatmap(corr_matrix,annot=True,cmap='viridis',fmt='.2f')
plt.title('Correlation matrix of investment across years')
plt.show()
```

